

SHIFT-SHARE DESIGNS

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- Setting: Observations $i = 1, \dots, N$ corresponding to regions (commuting zones (CZs), counties etc). Interested in effect of some treatment D_i on outcome Y_i .
- Construct regional instruments X_i by combining national-level *shifters* (or shocks) $\mathcal{X}_s$ to different sectors s (e.g. industries) with regional *shares* (or weights) w_{is} ,

$$X_i = \sum_{s=1}^S w_{is} \mathcal{X}_s, \quad w_{is} \geq 0.$$

So X_i is a *shift-share instrument*.

- Examples:
 - In Bartik (1991), want to estimate inverse labor supply elasticity: Y_i is log change in wages, and D_i is log change in employment. Need an instrument for labor demand. Construct instrument by combining is employment growth of industry s , $\mathcal{X}_s$, with lagged employment shares w_{is} . Similar setup in Blanchard and Katz (1992).
 - In Autor, Dorn, and Hanson (2013, ADH), interested in effect of Chinese import competition on US labor markets. Here Y_i is some local labor market outcome (employment, wages, etc), D_i measures the rise of Chinese imports to location i . Since weak labor markets may be more likely to import, use instrument where $\mathcal{X}_s$ is growth of Chinese exports to non-US countries in industry s , and w_{is} are lagged industry shares.
 - Many papers on effects of immigration (e.g. Card 2001). Interested in effect of local immigration D_i on wage gap Y_i between immigrant and native men. Here i can be a region-skill group cell; or an education-experience cell defined at a national level; or simply a region. As w_{is} use share of immigrants from country s , and $\mathcal{X}_s$ is a (normalized) change or growth in the number of immigrants from country s . Very similar to the logic of Bartik (1991).
 - Many other types of applications: effects on local labor markets of trade competition, technological change, or credit supply; impact of sectoral shocks on

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marriage patters, crime levels, innovation... See, e.g., introduction to Adão, Kolesár, and Morales (2019) for a partial list.

- How do we think about:
 - Identification?
 - Inference?

1. IDENTIFICATION

We need a sense in which X_i is as good as randomly assigned. Two approaches:

1. View w_{is} as randomly assigned. Explored in Goldsmith-Pinkham, Sorkin, and Swift (2020). They then view the setup as an overidentified instrumental variables model, with X_i as one of many possible ways of combining the “instruments” w_{is} .
 - If we believe exogeneity of w_{is} , no issues with estimation of inference, so long as (i) $n \rightarrow \infty$, and (ii) we can form groups of regions across which w_{is} are independent.
 - Can try to “open the black box” of the constructed instrument by looking at the implicit weights X_i puts on each share, called “Rotemberg weights” in this context; it’s an instrumental variables (IV) analog to leverage analysis in ordinary least squares (OLS).
 - But the shares w_{is} not plausibly exogenous in many applications: worry there are shift-share terms with structure similar to X_i in the structural residual (see Section 1.1 below).
2. View $\mathcal{X}_s$ as randomly assigned. First proposed by a working-paper version of Borusyak, Hull, and Jaravel (2022), and advocated for by Adão, Kolesár, and Morales (2019). Focus on here.

1.1. Stylized economic model

To serve as a background to help think through econometric issues, useful to consider a stylized model (taken from Adão, Kolesár, and Morales 2019).

SETUP

- Economy with multiple sectors $s = 1, \dots, S$ and regions $i = 1, \dots, J$
- Labor demand and supply depend on wage ω_i :

$$\begin{array}{ll} \text{Demand of } s \text{ in } i: & \log L_{is} = -\sigma_s \log \omega_i + \log D_{is}, \quad \sigma_s > 0, \\ \text{Supply in } i: & \log L_i = \phi \log \omega_i + \log S_i, \quad \phi > 0, \end{array}$$

σ_s and ϕ : labor demand and supply elasticities; S_i : supply shifter;

- Decompose labor demand shifter D_{is} into observed shifter of interest χ_s and potentially unobserved components:

$$\log D_{is} = \rho_s \log \chi_s + \log \mu_s + \log \eta_{is}.$$

- Also decompose labor supply shocks into group-specific shifters:

$$\log S_i = \sum_{g=1}^G \tilde{w}_{ig} \log v_g + \log v_i, \quad (1)$$

- Workers assumed immobile across regions, freely mobile across sectors $\implies$ market clearing: $L_i = \sum_{s=1}^S L_{is}$ for $i = 1, \dots, J$.
- Assume $\{\hat{\chi}_s, \hat{\mu}_s, \hat{\eta}_{is}, \hat{v}_g, \hat{v}_i\}_{i,s,g} \sim F$, where $\hat{z} = \log(z^t/z^0)$.

EQUILIBRIUM

- Up to first-order approximation around initial equilibrium

$$\hat{L}_i = \sum_{s=1}^S l_{is}^0 (\gamma_{is} \hat{\chi}_s + \lambda_i \hat{\mu}_s + \lambda_i \hat{\eta}_{is}) + (1 - \lambda_i) \left(\sum_{g=1}^G \tilde{w}_{ig} \hat{v}_g + \hat{v}_i \right),$$

l_{is}^0 : initial employment share, $\lambda_i \equiv \phi [\phi + \sum_s l_{is}^0 \sigma_s]^{-1}$, and $\gamma_{is} \equiv \rho_s \lambda_i$. Similar to a reduced-form regression considered in Autor, Dorn, and Hanson (2013).

- Solving for wages yields a regression of the sort considered in Bartik (1991) or Blanchard and Katz (1992)

$$\hat{\omega}_i = \frac{1}{\phi} \hat{L}_i - \frac{1}{\phi} \left(\sum_{g=1}^G \tilde{w}_{ig} \log v_g + \log v_i \right)$$

$\hat{L}_i$ is endogenous, but can use demand shifters $\hat{\chi}_s$ to estimate $1/\phi$ using IV, since $\phi^{-1} = \frac{\partial \hat{\omega}_i / \partial \hat{\chi}_s}{\partial \hat{L}_i / \partial \hat{\chi}_s}$

KEY TAKEAWAYS

1. Change in regional employment combines multiple shift-share terms; complicated correlation structure in residuals
2. Shifter effects depend on parameters that are heterogeneous across i and s .

1.2. Econometric model

To make identification arguments precise, focus on reduced-form effect of X_i onto Y_i for clarity of argument.

- As always, first ask: what do we mean by “effect”? What’s the ideal experiment here?
- Use potential outcomes, $Y_i(\chi_1, \dots, \chi_s) = Y_i(0) + \sum_s w_{is} \chi_s \beta_{is}$. Observed outcome is $Y_i(\mathcal{X}_1, \dots, \mathcal{X}_s)$. (exercise: how would one define potential outcomes in Goldsmith-Pinkham, Sorkin, and Swift (2020)? What’s the idealized experiment? Can we think of an economic model with such a structure?)
 - ADH example: What would regional outcomes be if we assign different shocks $\mathcal{X}_s$ to Chinese export growth? We posit here effect proportional to regional employment exposure to industry s , w_{is} .
 - For studying effect of demand shocks $\hat{\chi}_s$ in stylized model,

$$Y_i = \hat{L}_i, w_{is} = l_{is}^0, \chi_s = \hat{\chi}_s, \beta_{is} = \gamma_{is},$$

and $Y_i(0) = \lambda_i \sum_{s=1}^S w_{is} (\hat{\mu}_s + \hat{\eta}_{is}) + (1 - \lambda_i) (\sum_{g=1}^G \tilde{w}_{ig} \hat{v}_g + \hat{v}_i)$ aggregates all shifters other than shifter of interest $\hat{\chi}_s$.

- w_{is} are be equilibrium objects, condition on them throughout.

If there are no controls, then we need $\mathcal{X}$ to be as good as randomly assigned, in line with discussion in OLS section of the course: $E[\mathcal{X}_s | \mathcal{F}_0] = 0$, $\mathcal{F}_0 = (Y(0), B, W)$. We use matrix notation: A has N rows, $\mathcal{A}$ has S rows, B has elements β_{is} , W has elements w_{is} . OLS estimand:

$$\beta = \frac{\sum_i E[X_i Y_i | \mathcal{F}_0]}{\sum_i E[X_i^2 | \mathcal{F}_0]} = \frac{\sum_{i=1}^N \sum_{s=1}^S \pi_{is} \beta_{is}}{\sum_{i=1}^N \sum_{s=1}^S \pi_{is}}, \quad (2)$$

where $\pi_{is} = w_{is}^2 \text{var}(\mathcal{X}_s | W)$.

- Treats population of interest to be sample at hand. Answers the question of what we’d be estimating in an idealized experiment where $\mathcal{X}_s$ are randomly assigned.

CONTROLS If there are controls with exact shift-share structure, $Z_i = \sum_s w_{is} Z_s$, then need to assume (why?)

$$E[\mathcal{X} | \mathcal{Z}, U, Y(0), B, W] = \mathcal{Z}' \gamma.$$

- Important: this implies that if $\sum_s w_{is} \neq 1$, need to include $\sum_s w_{is}$ as control (why?)

Then the weights in eq. (2) become $\pi_{is} = w_{is}^2 \text{var}(\mathcal{X}_s | W, \mathcal{Z})$, and X_i is replaced by residual after projecting it off Z_i .

- What if controls don’t have an exact shift-share structure? One option is to assume that such controls “proxy” for unobserved sectoral shocks, see Adão, Kolesár, and

Morales (2019) for a formalization of this idea. **Research question:** is this the best way of thinking about the issue? Can we give some other explanation for shift-share regressions are not run at a sectoral level?

Concerns raised in the literature we did not address:

1. Is β an interesting object? Is it policy relevant?
2. What are we estimating in the presence of cross-regional spillovers?

2. ESTIMATION AND INFERENCE

Notice that in stylized economic model, $Y_i(0)$ (and hence regression residual) incorporates terms that have shift-share structure, with shares that could be identical to (other demand shocks μ_s) or different from (supply shocks to occupational groups, v_g) w_{is} . Correct inference requires taking into account potential cross-regional correlation in residuals across observations with similar values of w_{is} .

2.1. Placebo evidence

- Use 2000–2007 observed changes in labor market outcomes $\{Y_i\}_{i=1}^n$ and 1990 employment shares $\{w_{is}\}_{i=1, s=1}^{N, S}$ for US commuting zones.
- $N = 722$ regions, and $S = 397$ sectors corresponding to 4-digit Standard Industrial Classification (SIC) codes
- For each simulation draw m , generate $\mathcal{X}_s^m \sim \mathcal{N}(0, 5)$, and estimate

$$Y_i = \delta + \beta \sum_{s=1}^S w_{is} \mathcal{X}_s^m + \epsilon_i.$$

Computer-generated shocks cannot have impacted US labor mkt outcomes $\implies \beta = 0$.

- Consider Eicker-White (EHW) (*Robust*) and state-clustered standard errors (*Cluster*), commonly used in practice.

Results reported in Table 1, taken from Adão, Kolesár, and Morales (2019). In line with previous discussion: regions with similar sectoral employment shares $\{w_{is}\}_{s=1}^S$ also tend to have similar regression residuals and hence similar $X_i \epsilon_i$. Correlation independent of regions' geographic location $\implies$ not captured by clustering on geographic units.

Table 1: Standard errors and rejection rate of the hypothesis $H_0: \beta = 0$ at 5% significance level.

	Estimate		Median std. error		Rejection rate	
	Mean (1)	SD (2)	Robust (3)	Cluster (4)	Robust (5)	Cluster (6)
Panel A: Change in the share of working-age population						
Employed	−0.01	2.00	0.73	0.92	48.5%	38.1%
Employed in mfg	−0.01	1.88	0.60	0.76	55.7%	44.8%
Employed in non-mfg	0.00	0.94	0.58	0.67	23.2%	17.6%
Panel B: Change in average log weekly wage						
Employed	−0.03	2.66	1.01	1.33	47.3%	34.2%
Employed in mfg	−0.03	2.92	1.68	2.11	26.7%	16.8%
Employed in non-mfg	−0.02	2.64	1.05	1.33	45.4%	33.7%

Notes: For the outcome variable indicated in the leftmost column, this table indicates the mean and standard deviation of the OLS estimates of across the placebo samples (columns (1) and (2)), the median standard error estimates (columns (3) and (4)), and the percentage of placebo samples for which we reject the null hypothesis $H_0: \beta = 0$ using a 5% significance level test (columns (5) and (6)). *Robust* is the Eicker-Huber-White standard error, and *Cluster* is the standard error that clusters CZs in the same state. Results are based on 30,000 placebo samples.

2.2. Consistency and correct standard errors

To help focus on key issues, assume $\beta_{is} = \beta$, and abstract from controls. See Adão, Kolesár, and Morales (2019) for results with controls. Then

$$\hat{\beta} - \beta = \frac{\sum_i X_i Y_i(0)}{\sum_i X_i^2} = \frac{\sum_s \mathcal{X}_s \sum_i w_{is} Y_i(0)}{\sum_{s,t} \mathcal{X}_s \mathcal{X}_t \sum_i w_{is} w_{it}}.$$

Use randomization-style framework where we condition on $\mathcal{F}_0 = (Y(0), W)$, and consider repeated sampling of $\mathcal{X}_s$. This is the implicit framework used in the placebo. Would want only 5 of 100 researchers find a non-zero treatment effect if each of them comes up with a different set of irrelevant shifters.

Key substantive assumption: $\mathcal{X}_s$ independent across s (but perhaps not identically distributed). No restriction on correlation structure of objects in $\mathcal{F}_0$.

- Need $S \rightarrow \infty$ for consistency.
- To ensure low “leverage”, also need sectors to be small in the sense that $\max_s n_s / N \rightarrow 0$, where $n_s = \sum_i w_{is}$ is total share of sector s (and also regularity conditions on $Y_i(0)$)

Intuition: consider “concentrated sectors”, $w_{is} = \mathbb{1}\{s = s(i)\}$, where $s(i)$ is concentration of region i . Then setup equivalent to OLS in randomized controlled trials with cluster-level treatment, assumption equivalent to the largest cluster asymptotically negligible.

- In contrast, Goldsmith-Pinkham, Sorkin, and Swift (2020) can live with small, finite S .

Standard error (call it AKM): corresponds to standard deviation of $\sum_i X_i Y_i(0)$ conditional $\mathcal{F}_0$, divided by $\sum_i X_i^2$: usual sandwich form with bread as usual, but different meat today:

$$\widehat{se}(\hat{\beta}) = \frac{\sqrt{\sum_{s=1}^S \mathcal{X}_s^2 \hat{R}_s^2}}{\sum_{i=1}^N X_i^2}, \quad \hat{R}_s = \sum_{i=1}^N w_{is} \hat{\epsilon}_i.$$

With “concentrated sectors”, $w_{is} = \mathbb{1}\{s = s(i)\}$, $\widehat{se}(\hat{\beta})$ equivalent to clustering regions that specialize in same sector—very different from clustering on state!

- In line with rule of thumb that one should “cluster” at level of variation of regressor of interest.
- Formula essentially forms sectoral clusters, using w_{is} to weight residuals.
- In practice, to help mitigate small-sample issues arising from the fact that the effective number of clusters/sectors may be small, it helps to impose the null when estimating the residual; standard error then computed by test inversion (call it AKMo). Closed-form formula from solving a quadratic equation. Tends to improve finite-sample coverage, and longer CI length. AKM vs AKMo analogous to Wald vs LM tests in likelihood models. In IV context, AKMo generalizes the Anderson and Rubin (1949) CI to allow structural errors to be correlated: robust to weak instruments.

3. EMPIRICAL APPLICATIONS

3.1. ADH

- Interested in effect of Chinese exports on US labor mkt outcomes. i : CZ; s : 4-digit SIC code, as in placebo. Use 1990–2007 data (2-period panel).
- Estimate shift-share IV regression with:
 - Outcome: 10-year equivalent change in labor mkt outcomes
 - Endogenous variable: shift-share regressor, Chinese exports to US aggregated using beginning-of-period employment shares
 - Shift-share instrument: Chinese exports to rest of world, aggregated using beginning-of-period employment shares
 - Largest set of controls used in ADH
- Results in Table 2. CIs on average 23% (AKM) or 66% (AKMo) wider, significance not affected

Table 2: Effect of Chinese exports on Change in the employment shares in U.S. commuting zones—Autor, Dorn, and Hanson (2013)

	All (1)	Manuf. (2)	Non-Manuf. (3)
Panel A: IV Regression			
$\hat{\beta}$	−0.77	−0.60	−0.18
Robust	[−1.10, −0.45]	[−0.78, −0.41]	[−0.47, 0.12]
Cluster	[−1.12, −0.42]	[−0.79, −0.40]	[−0.45, 0.10]
AKM	[−1.25, −0.30]	[−0.84, −0.35]	[−0.54, 0.18]
AKMo	[−1.69, −0.39]	[−1.01, −0.36]	[−0.84, 0.14]
Panel B: OLS Reduced-Form Regression			
$\hat{\beta}$	−0.49	−0.38	−0.11
Robust	[−0.71, −0.27]	[−0.48, −0.28]	[−0.31, 0.08]
Cluster	[−0.64, −0.34]	[−0.45, −0.30]	[−0.27, 0.05]
AKM	[−0.81, −0.17]	[−0.52, −0.23]	[−0.35, 0.12]
AKMo	[−1.24, −0.24]	[−0.67, −0.25]	[−0.64, 0.08]

Notes: $N = 1,444$ (722 CZs $\times$ 2 time periods). Observations are weighted by the start of period CZ share of national population. All regressions include the full vector of baseline controls in ADH. 95% confidence intervals are reported in square brackets. AKM and AKMo methods cluster at 3-digit SIC level.

3.2. Labor supply elasticity

- Estimate $1/\phi$ in

$$\hat{\omega}_i = \frac{1}{\phi} \hat{L}_i + Z_i \delta + \epsilon_i$$

Z_i : same vector of controls as in ADH

- To instrument for $\hat{L}_i$, use
 1. National employment growth, as in Bartik (1991) (actually, a leave-one-out version of it); or
 2. Increase Chinese imports to high-income countries excl. US, as in ADH
- Results reported in Table 3. Two approaches give similar point estimates, but while AKM CIs broadly similar to usual ones with Bartik IV, 20% and 250% wider for AKM and AKM0 with ADH IV
 - National employment growth absorbs most sector-level shocks, not much shift-share structure left in residual

4. OPEN QUESTIONS

I am not clear on the econometrics in the following settings:

Table 3: Estimation of inverse labor supply elasticity

Dependent variable:	First-Stage $\hat{L}_i$ (1)	Reduced-Form $\hat{w}_i$ (2)	2SLS $\hat{w}_i$ (3)
Panel A: Bartik IV—Leave-one-out estimator			
$\hat{\beta}$	0.87	0.71	0.82
Robust	[0.68, 1.06]	[0.53, 0.89]	[0.65, 0.98]
Cluster	[0.62, 1.12]	[0.46, 0.96]	[0.60, 1.03]
AKM (<i>leave-one-out</i>)	[0.59, 1.15]	[0.47, 0.94]	[0.61, 1.02]
AKMo (<i>leave-one-out</i>)	[0.53, 1.15]	[0.42, 0.94]	[0.59, 1.09]
Panel B: ADH IV			
$\hat{\beta}$	−0.72	−0.48	0.67
Robust	[−1.04, −0.39]	[−0.80, −0.16]	[0.36, 0.98]
Cluster	[−0.93, −0.50]	[−0.78, −0.18]	[0.35, 0.99]
AKM	[−1.19, −0.24]	[−0.88, −0.07]	[0.27, 1.07]
AKMo	[−1.83, −0.35]	[−1.27, −0.10]	[0.18, 1.14]

Notes: $N = 1,444$ (722 CZs $\times$ 2 time periods). The variable $\hat{L}_i$ denotes the log-change in the employment rate in CZ i . The variable $\hat{w}_i$ denotes the log change in mean weekly earnings. Observations are weighted by the start of period CZ share of national population. All regressions include the full vector of baseline controls in ADH. 95% confidence intervals in square brackets. AKM and AKMo methods cluster at 3-digit SIC level.

- Some papers use IV regressions where the instrument is an interaction term, $X_{it} = X_t W_i$. See, for instance Nunn and Qian (2014), or Kearney and Levine (2015).
- In some settings, the number of sectors is as small as $S = 2$ (Nakamura and Steinsson 2014).
- There are a number of structural papers using shift-share instruments (e.g. Diamond 2016). Borusyak and Hull have a working paper thinking about instruments of the form $f_i(\mathcal{X}, w)$, but more work needs to be done here.

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